

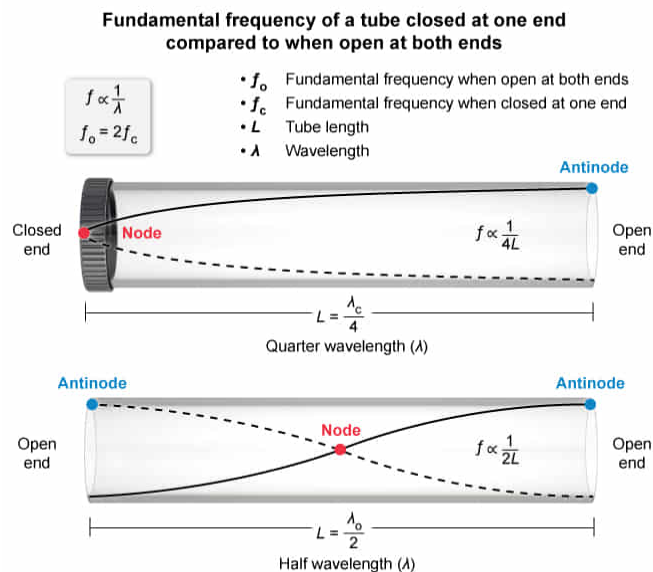
The fundamental frequency of a glass tube is measured to be f when the tube is capped at one end, but open at the other end. What is the fundamental frequency of the tube when the end cap is removed and both ends are open?

- ☐ A. $f/4$ [8%]
☐ B. $f/2$ [37%]
☐ C. f [5%]
✔ ☒ D. $2f$ [48%]

Correct

48% answered correctly

Explanation:



Standing sound waves are generated in tubes due to interference between incident waves and reflected waves, which move in opposite directions in the air tube. The resonating sound waves in a pipe create nodes (ie, points of zero amplitude) at closed ends due to restricted air movement and antinodes (ie, points of maximal amplitude) at open ends due to free motion of air molecules.

A **pipe open on both ends** will resonate at the fundamental frequency when antinodes are formed at each end. A pipe with **one end open** will resonate at the fundamental frequency when a node is formed at the closed end and an antinode is formed at the open end.

In this question, the glass tube is initially closed c at one end and then open o

at both ends. At the fundamental frequency, the wavelengths of sound are twice and four times the tube length L :

$$\lambda_o = 2L$$

$$\lambda_c = 4L$$

Therefore,

$$\lambda_c = 2\lambda_o$$

The wave speed v is the product of the wavelength λ and **frequency** f of a **wave**:

$$v = \lambda f$$

Because the speed of sound is constant in the tube:

$$v_o = v_c$$

$$f_o \lambda_o = f_c \lambda_c$$

$$f_o \cancel{\lambda_o} = f_c (2\cancel{\lambda_o})$$

$$f_o = 2f_c$$

Therefore, the fundamental frequency of the glass tube when open at both ends is *twice* that of the fundamental frequency when the tube is closed at one end.

(Choice A) $f/4$ is the result of an error made by plugging in the wavelength for a closed-end tube into wave speed equation ($f = v/4L$).

(Choice B) $f/2$ would be correct if the question instead asked for the fundamental frequency of the tube closed at one end.

(Choice C) f would be true if the tube length changed to maintain the same fundamental frequency.

Educational objective:

The fundamental frequency of standing sound waves in a tube of fixed length depends on whether the pipe is open at both ends or closed at one end.

Time spent: 0 sec

Subject:
Physics

QID: 404082

System:
Light and Sound

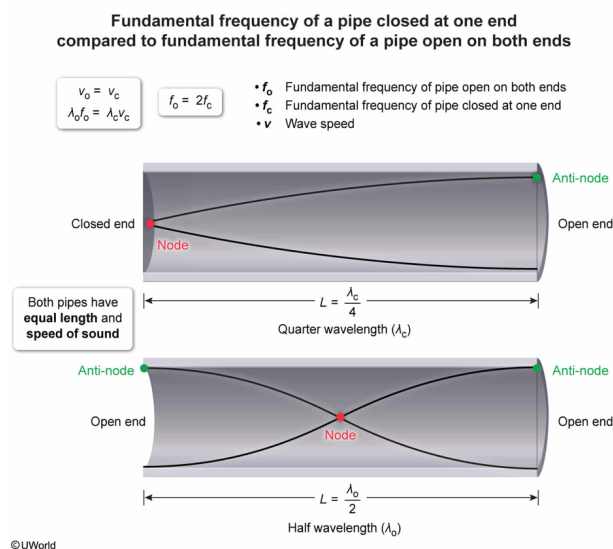
A speaker and frequency generator are used to create resonances in two pipes. Pipe A of length L is open at both ends. Pipe B is also of length L but is closed at one end. Which statement about the fundamental frequencies produced by the pipes is true?

- ✔ ☒ A. The fundamental frequency of pipe A will be twice as great as the fundamental frequency of B, because antinodes must occur at open ends and nodes must occur at closed ends. [60%]
- ☐ B. The fundamental frequency of pipe A will be half as great as the fundamental frequency of B, because antinodes must occur at open ends and nodes must occur at closed ends. [33%]
- ☐ C. The fundamental frequencies of the pipes are the same, because the speed of sound is the same in both pipes. [2%]
- ☐ D. The fundamental frequencies of the pipes are the same, because all pipes of equal length will resonate at the same frequency. [3%]

Correct

59% answered correctly

Explanation:



Standing sound waves are generated in pipes due to interference between incident waves and reflected waves, which move in opposite directions in the air tube. The resonating sound waves in a pipe create nodes (points of zero amplitude) at closed ends due to restricted air movement, whereas at open ends the sound waves create antinodes (points of maximal amplitude) due to the free motion of air molecules.

A **pipe open on both ends** will resonate when antinodes are formed at each end. A pipe with **one end open** will resonate when a node is formed at the closed end and an antinode is formed at the open end.

In this question, there are two pipes of equal length L ; pipe A is open (o) at both ends and pipe B is closed (c) at one end:

$$L_o = L_c = L$$

The speed of sound v is also equal in both pipes:

$$v_o = v_c$$

Because v is the product of **frequency and wavelength**:

$$\lambda_o f_o = \lambda_c f_c$$

$$f_o = \frac{\lambda_c f_c}{\lambda_o}$$

At the fundamental frequency, the wavelengths of the sound are related to the lengths of the pipes:

$$L = \frac{\lambda_o}{2} L = \frac{\lambda_c}{4}$$
$$\lambda_o = 2L \quad \lambda_c = 4L$$

Consequently, using the above equation for f_o :

$$f_o = \frac{4L f_c}{2L}$$
$$f_o = 2f_c$$

Therefore, pipe A 's fundamental frequency will be *twice* as great as the fundamental frequency of pipe B .

(Choice B) Antinodes occur at open ends and nodes at closed ends. However, this implies pipe A has a higher fundamental frequency than pipe B .

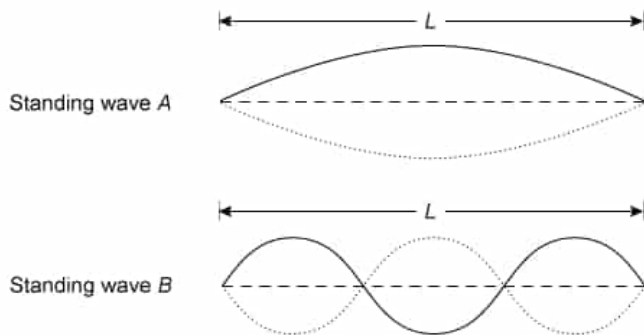
(Choice C) The speed of sound for both pipes is equal, but pipe A has a higher fundamental frequency.

(Choice D) Although the pipes are of equal length, the fundamental frequency is determined by whether the pipe ends are open or closed. Here, pipe A has twice the fundamental frequency of pipe B .

Educational objective:

The fundamental frequencies of standing waves in open and closed pipes of equal length are different. Antinodes occur at open ends and nodes occur at closed ends.

Standing wave *A* is produced by vibrating a string that is fixed at both ends with a frequency f . A second standing wave *B* is produced on the same string, as shown in the figure below. Which of the following statements about standing waves *B* and *A* is true?



- ☐ A. Because wave *B* occurs on a string that is the same length as wave *A*, the frequency that causes resonance for wave *B* is the same as f . [5%]
- ☐ B. Because the wavelength of *B* is less than the wavelength of *A*, the speed of the waves along the string for wave *B* is less than the speed for wave *A*. [9%]
- ☐ C. Because the frequency of wave *B* is greater than the frequency of wave *A*, the speed of the waves along the string for wave *B* is greater than the speed for wave *A*. [51%]
- ✔ ☒ D. Because resonance occurs only at an integer multiple of the fundamental frequency for a string, standing wave *B* occurs at a frequency $3f$. [33%]

Correct

34% answered correctly

Explanation:

The harmonic (resonance) frequencies for a standing wave on a string

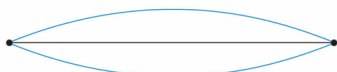
$$\lambda_n = \frac{2}{n} L$$

$$f_n = n f_1$$

- λ Wavelength
- L Length of string
- n Integer multiple ($n = 1, 2, 3, \dots$)
- f Frequency

Fundamental frequency

f_1

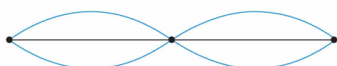


Standing wave A

$$\lambda_1 = 2L$$

Second harmonic

$f_2 = 2f_1$



Third harmonic

$f_3 = 3f_1$



Standing wave B

$$\lambda_3 = \frac{2}{3} L$$

L

Resonance generates **standing waves** on a string fixed at both ends due to interference between incident waves and reflected waves, which move in opposite directions along the string. The resonating standing waves create nodes (points of zero amplitude) and antinodes (points of maximal

amplitude). At the **fundamental resonance**, there is one antinode and the **wavelength λ** of the standing wave is twice the length of the string L :

$$\lambda = 2L$$

The wavelengths of the higher order harmonics generated on a string are smaller, occurring at integer factors of the string length:

$$\lambda_n = \frac{2}{n}L$$

Because λ equals wave speed v divided by the frequency f , λ is **inversely proportional** to f :

$$\lambda = \frac{v}{f}$$

Hence, subsequent resonances occur at an integer multiple n ($n = 1, 2, 3, \dots$) of the fundamental frequency:

$$f_n = nf_1$$

In this question, the wavelength of standing wave A is twice the length of the string L . Hence, the **fundamental resonance frequency f_1** occurs on the string in standing wave A :

$$\lambda = 2L$$

For standing wave B , the wavelength on the string is equal to two-thirds the length of the string:

$$\lambda = \frac{2}{3}L$$

Hence, $n = 3$ and the third harmonic resonance occurs for standing wave B . Therefore, the third resonance occurs when the frequency is three times the fundamental resonance frequency f_1 :

$$f_3 = 3f_1$$

Choice A) The length of the string is the same for both waves. However, wavelength decreases with each standing wave and is $\frac{2L}{3}$ for the third resonance.

(Choices B and C) Although the wavelength of wave B is less than that of wave A and the frequency of wave B is greater than that of wave A , the speed of both waves must be equal because the medium (string) is the same.

Educational objective:

Resonance generates standing waves in a medium. The frequency of standing waves on a string occurs at an integer multiple of the fundamental resonance frequency.

Time spent:0 sec

Subject:
Physics

QID:404084

System:
Light and Sound

The air in an adjustable pipe open at one end and closed at the other resonates at the fundamental frequency when the pipe has length L_0 . The pipe is then adjusted such that it is twice its original length. What is the wavelength of the air in the pipe at the new fundamental frequency?

- ☐ A. L_0 [6%]
- ☐ B. $2L_0$ [30%]
- ☐ C. $4L_0$ [26%]
- ✓ ☒ D. $8L_0$ [36%]

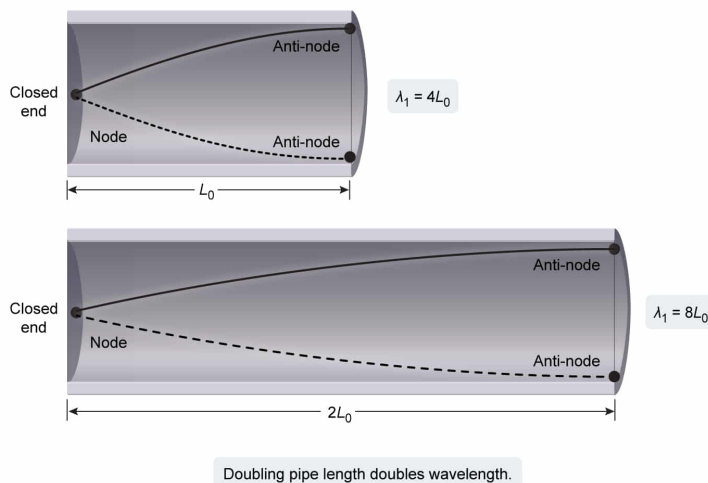
Correct

35% answered correctly

Explanation:

Wavelength of sound at the fundamental frequency in a pipe closed at one end

- λ_1 Wavelength at fundamental frequency
- L_0 Length of pipe



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Standing waves (ie, a wave pattern that appears to stand in place) form due to the superposition of incoming and reflected waves in a region with confined boundaries. Examples of standing waves are the resonances of sound waves in the air contained in a pipe or a tube. The **wavelength (λ)** and **frequency (f)** of a standing wave are determined by the speed of the waves and the size of the region.

For a pipe with **one end closed**, the air molecules at the open end are free to vibrate, producing an antinode, but at the closed end the molecules are fixed to have zero displacement (ie, a node). Hence, for pipe length L it follows that the allowed wavelengths λ_n of the standing waves are:

$$\lambda_n = \frac{4L}{n}$$

where n is an odd integer.

In this question, the air in the pipe vibrates at the fundamental frequency ($n = 1$) when pipe length $L = L_0$. Consequently, the wavelength of the air is initially:

$$\lambda_1 = 4L_0$$

The pipe is then adjusted to a new length L' twice its initial length:

$$L' = 2L_0$$

Because the wavelength of the standing wave in the fundamental frequency must always be four times the pipe length, it follows that the new wavelength λ'_1 is:

$$\lambda'_1 = 4L'$$

$$\lambda'_1 = 8L_0$$

(Choices A and B) The wavelength of the standing wave in air is initially $4L_0$. Increasing the pipe length cannot decrease the wavelength at the fundamental frequency.

(Choice C) $4L_0$ would be the wavelength of the standing wave if the pipe had **both ends open**.

Educational objective:

A pipe with one end closed resonates when a node occurs at the

closed end and an antinode occurs at the open end. The wavelength of each resonance is determined by the length of the pipe.

Time spent:0 sec

Subject:
Physics

QID:404085

System:
Light and Sound